YUANKAI TAO

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EDUCATION

Georgia Institute of Technology *Aug 2026 — Dec 2027*
M.S. in Operations Research, Stewart School of Industrial and Systems Engineering
Selected Coursework: Deterministic Optimization, Computational Statistics, Simulation

University of California, San Diego *Sep 2024 — Dec 2025*
M.S. in Quantitative Finance, Rady School of Management *GPA:* 3.84/4.0
Selected Coursework: Data Science for Finance, Computational Finance Methods, Deep Learning

Zhongnan University of Economics and Law *Sep 2020 — Jun 2024*
B.S. in Management, School of Accounting and Finance *GPA:* 3.63/4.0
Selected Coursework: Probability & Statistics, Accounting, Linear Algebra, Python Programming

TECHNICAL SKILLS

Analytics & Reporting	**SQL, Excel, VBA**, Data Visualization, Dashboarding, KPI Reporting
Business Analysis	Requirements Analysis, Root-Cause Analysis, Data Reconciliation
Programming & Tools	Excel, Word, PowerPoint, SQL, Python, Bloomberg, Git, Linux
Libraries & Methods	Pandas, NumPy, Matplotlib, Statistical Analysis, Forecasting, Web Scraping

PROFESSIONAL EXPERIENCE

VariFlight Technology *Jul 2025 — Sep 2025*
Data Analyst, Intern *Data Development Department*

• Built Python data-collection workflows to aggregate public reports, webpages, and images from the Civil Aviation Administration of China, creating structured inputs for recurring analysis and reporting.

• Reconciled unstructured information across **600+ PDF reports and 500+ webpages**; used OCR and table-extraction tools to standardize raw data into analysis-ready files, improving data consistency and repeatability.

• Created dashboards for key aviation metrics, including transportation and passenger turnover, translating operational data into clear trends and actionable insights for internal users.

Haitong Securities *Jan 2024 — May 2024*
Investment Analyst, Intern *Investment Banking Department*

• Conducted IPO due diligence using Wind financial terminal; validated client creditworthiness and investigated material fluctuations in sales and procurement data to identify exceptions and potential business risks.

• Benchmarked R&D and other key expenditures across peer companies, summarized variance drivers, and developed preliminary recommendations to support IPO filing priorities and management review.

Ernst & Young *Dec 2022 — Feb 2023*
Audit, Intern *Financial Service Office (FSO)*

• Analyzed intercompany and investment transactions for Hyundai Motor Finance Company; performed variance analysis, investigated significant deviations, reviewed supporting records, and documented findings and recommendations.

• Automated the cleaning and consolidation of financial tables from **220+ CSV files**, producing standardized audit outputs and improving the efficiency and consistency of recurring reporting.

Guangdong Hedge Fund Investment Management Co., Ltd. *May 2026 — Jul 2026*
Quantitative Research, Intern *Quantitative Strategy Department*

• Integrated market data and 16 indicators covering trend, volatility, liquidity, volume, and open interest into a repeatable Python workflow for performance, risk, and transaction-cost reporting.

SELECTED PROJECTS

Text Analysis and BERT Fine-tuning on SEC 10-K Reports *MGTF 423, UCSD*
Team Lead | Analysis of 20.5M Text Rows

• Analyzed SEC filings using named-entity recognition, frequency profiling, and word clouds to surface patterns in company disclosures and communicate findings visually.

• Led an iterative human-in-the-loop workflow with three annotation rounds; improved scarce-label data and delivered a model with **79% accuracy** for measuring uncertainty tone.

UCSD / FMSbonds, Inc. 2025 Capstone Project *Aug 2025 — Dec 2025*
Team Lead | Bond Pricing Process Optimization

• Mapped high-frequency cash-flow queries and identified redundant object generation as a process bottleneck; implemented caching that reduced per-bond runtime from 1.17 ms to 0.13 ms (**9× faster**).

• Redesigned the yield-to-worst calculation workflow using Numba JIT, reducing runtime from 1.57 ms to 7.5 μs per bond (**210× faster**) and enabling scalable portfolio reporting.

RESEARCH EXPERIENCE

Research Assistant for Professor Franklin Qian, UNC-Chapel Hill *Mar 2025 — Present*
The Numbers Game: Signaling Effects of Pricing Formats in Housing Bargaining

• Built a Python and SQL data pipeline combining a **76-million-observation spatial dataset** with property and population data across multiple sources on Google Cloud Platform.

• Defined property-level filtering rules and validation checks for comparable-housing analysis; parallelized recurring calculations to achieve approximately **50× computational speedup**.

• Managed and queried a **964.5M-record, 100GB+ database** using PostgreSQL, PostGIS, BigQuery, and DuckDB, delivering reproducible datasets for research analysis.

CERTIFICATIONS & HONORS

Algorithms and Data Structures Certificate (6-Course Certificate) — Coursera, UC San Diego
Deep Learning Specialization (5-Course Certificate) — Coursera, DeepLearning.AI
Beta Gamma Sigma (Top 20% of graduating class) — UCSD, *2024 — 2025*
WorldQuant Alphathon 2024, Gold Level (Rank 16, Top 3.1%) — *2024*
Special Scholarship in Innovation and Entrepreneurship — ZUEL, *2022 — 2023*